

Week 4: Probability of Events and Decision-Making

Date: September 21, 2026

Reference: Statistics in Everyday Life Using Excel (Ahn Gi-su, 2021).

Some image sources: Chunjae Education, Probability and Statistics (Ryu Hee-chan).

Learning Objectives

- Should we make this investment?
- Expected return and risk
- Random variables and probability distributions

Recent KOSPI Trend

Since the June 2026 peak, the market has repeatedly fallen sharply and rebounded.

Date	Closing level	Description
2026-06-22	9,114.55	Record-high close
2026-07-08	7,246.79	Down 5.35% in one day; sell-side trading curb triggered
2026-09-18	6,894.23	Rebounded 2.66% in one day

The September 18 close was about 24.4% below the peak. 'The price has fallen a lot!' Does that mean we should buy?

Source: DigitalToday, Yahoo Finance, and Seoul Shinmun; checked September 20, 2026. Figures are based on trading days.

Reference: What Is the KOSPI?

KOSPI stands for Korea Composite Stock Price Index.

- It is a major stock index that tracks changes in the total market capitalization of companies listed on the KOSPI market.
- It uses January 4, 1980 as the base date, with the index set to 100.
- A rising index is interpreted as an increase in the overall market value; a falling index indicates a decrease.
- Large companies such as Samsung Electronics and SK hynix have a major effect on the index.

Saying that 'the KOSPI rose or fell' summarizes the movement of the entire Korean stock market in a single number.

Source: KRX index description (base date: 1980-01-04 = 100).

If Stock Prices Have Fallen Sharply, Are They 'Cheap'?

Investment decisions should consider not only the price level, but also earnings, assets, and uncertainty in returns.

1. Price

The KOSPI has fallen substantially from its peak, so it may look cheaper.

2. Value

Compare the current price with the company's earnings and net assets. Measures include PER, PBR, and ROE.

3. Risk

Even when two investments have the same expected return, the amount by which their outcomes vary can differ. The slide leaves two risk measures blank for discussion.

Is an investment automatically good if its expected return is positive?

Reference: $PBR = ROE \times PER$

PER (Price-to-Earnings Ratio): stock price relative to earnings

ROE (Return on Equity): net income relative to shareholders' equity

PBR (Price-to-Book Ratio): stock price relative to book value

$\text{Market capitalization} / \text{Net assets} = (\text{Net income} / \text{Net assets}) \times (\text{Market capitalization} / \text{Net income})$

In the slide's simplified explanation, increasing net income and reducing net assets raises the ratios associated with market capitalization or stock price.

Reference: PER and PBR

1. PER (Price-to-Earnings Ratio)

'How many times the company's earnings is the stock price?'

Example: One share costs KRW 1,000, and the company earns KRW 100 per share each year. The stock is valued at 10 times its earnings, so $PER = 10$.

2. PBR (Price-to-Book Ratio)

'How many times the company's net assets - land, buildings, cash, and so on - is the value of its stock?'

Example: A bakery owns machinery and a store worth KRW 100 million in net assets. If buying the bakery costs KRW 200 million, $PBR = 2$. It is being sold for twice the value of its net assets.

Should We Make This Investment?

Situation: Assume there are only two possible investment outcomes.

- 90% probability: loss of KRW 1,000,000
- 10% probability: profit of KRW 10,000,000

1. Calculate the expected return.

2. Calculate the risk. (The slide presents this as the next step.)

Arithmetic Mean vs. Expected Value

Arithmetic mean: Add the values and divide by the number of values.

Expected value (expected return / weighted mean): Multiply each value by its probability (weight), then add the products.

1. Coin game with equal probabilities

Heads: gain KRW 200. Tails: lose KRW 100.

Arithmetic mean: $(200 + (-100)) / 2 = 50$

Expected value: $0.5(200) + 0.5(-100) = 50$

When probabilities are equal, the arithmetic mean equals the expected value.

3. Unequal probabilities

90% chance of -KRW 100; 10% chance of +KRW 1,000.

Arithmetic mean: $(-100 + 1,000) / 2 = 450$

Expected value: $0.9(-100) + 0.1(1,000) = 10$

This is a large difference.

Event Probabilities Are Often Unequal

In many everyday situations, the probabilities of different events are not equal.

Expected value = expected return = weighted mean = a type of average

Should We Make This Investment?

Situation: There are only two possible outcomes.

- 90% probability: loss of KRW 1,000,000
- 10% probability: profit of KRW 10,000,000

1. Expected return

$0.9(-100) + 0.1(1,000) = 10$, measured in units of KRW 10,000.

Therefore, the expected return is KRW 100,000.

2. Risk calculation

The next step is to measure how widely the possible outcomes vary around the expected return.

Expected Value and Variance of a Discrete Random Variable

Expected value

If a discrete random variable X has values $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$, then:

$$\mathbf{E(X) = x_1p_1 + x_2p_2 + \dots + x_np_n = \sum x_i p_i}$$

This is called the expected value or mean of X and is written as $E(X)$.

Variance and standard deviation

Let $m = E(X)$.

$$\mathbf{Variance: V(X) = E[(X - m)^2] = E(X^2) - [E(X)]^2}$$

$$\mathbf{Standard\ deviation: \sigma(X) = \sqrt{V(X)}}$$

Why Is Variance $E(X^2) - [E(X)]^2$?

Let $\mu = E(X)$. Expanding the definition of variance naturally produces the computational formula.

1. Start from the definition

$$\text{Var}(X) = E[(X - \mu)^2]$$

$$\mu = E(X)$$

$$\text{Deviation} = X - \mu$$

Variance = mean of squared deviations

2. Expand the square

$$E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2]$$

$$= E(X^2) - 2\mu E(X) + \mu^2$$

$$= E(X^2) - 2\mu^2 + \mu^2$$

$$= E(X^2) - \mu^2$$

3. Rearrange as a calculation formula

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Variance = 'mean of the squares' - 'square of the mean.'

Lesson points: (1) Variance is the mean of squared deviations. (2) In actual calculations, $E(X^2) - [E(X)]^2$ is more convenient. The slide repeats this second point as item 3.

Example 1: Expected Value and Variance of a Discrete Random Variable

A company produces 100 computer memory units per day. The number of defective units varies from day to day. When observing the number of defective units over 100 days, the number being studied is a random variable.

Date	Day 1	Day 2	Day 3	...	Day 97	Day 98	Day 99	Day 100
Defective units	2	4	0	...	2	3	2	1

Defective units X	Frequency	Relative frequency / Probability
0	5	$5/100 = 0.05$
1	25	$25/100 = 0.25$
2	40	$40/100 = 0.40$
3	25	$25/100 = 0.25$
4	5	$5/100 = 0.05$
Total	100	1

Using Table 1, we can calculate the frequency distribution and relative frequencies shown in Table 2. Fill the cells in the 'Example 1' worksheet and calculate the expected value and variance.

Example 2: Expected Return and Risk

The table pairs possible economic conditions with their probabilities to form a probability distribution of returns. Decide rationally whether to choose Investment A or Investment B.

Condition	Probability	Return on A	Return on B
Recession	0.25	-KRW 5 million	-KRW 1 million
Normal	0.50	KRW 2 million	KRW 2.5 million
Boom	0.25	KRW 10 million	KRW 5 million

Investment A

$E(X) = 0.25(-5) + 0.50(2) + 0.25(10) = \text{KRW } 2.25 \text{ million}$

Variance formula: $0.25(-5)^2 + 0.50(2)^2 + 0.25(10)^2 - (2.25)^2$

Investment B

$E(X) = 0.25(-1) + 0.50(2.5) + 0.25(5) = \text{KRW } 2.25 \text{ million}$

Variance formula: $0.25(-1)^2 + 0.50(2.5)^2 + 0.25(5)^2 - (2.25)^2$

Both investments have the same expected return, but Investment B has the lower variance and therefore lower risk. Monetary scaling affects the numerical variance unit; compare the two variances using the same unit.

Practice

Seongseo Company must choose one of two investments. Given the probability distributions below, fill in the cells in the 'Example 2' worksheet and decide which investment to choose.

Condition	Probability	Return on A	Return on B
Recession	0.3	-KRW 4 million	-KRW 2 million
Normal	0.5	KRW 2.5 million	KRW 3 million
Boom	0.2	KRW 10 million	KRW 7 million

Aggressive investors tend to choose the investment with the higher expected value, even if the difference is small. Conservative investors tend to choose the investment with lower risk. In this exercise, variance represents risk.

Types of Random Variables

Random variables are classified as discrete random variables and continuous random variables.

A variable with a finite or countable set of possible values is discrete. A variable that can take infinitely many values over an interval is continuous.

Type	Example	Possible values
Discrete	Number shown on one roll of a die	1, 2, 3, 4, 5, 6
Discrete	Number of employees at a company	0, 1, 2, 3, 4, ...
Discrete	Number of heads in two coin tosses	0, 1, 2
Continuous	Mobile phone battery usage time	$0 \leq X \leq 18$
Continuous	Students' body weight	$35 \leq X \leq 120$
Continuous	Monthly living expenses	$30,000 \leq X \leq 250,000$

Probability Distributions and Random Variables

A company produces 100 computer memory units per day, and the number of defective units changes daily. When the number of defects is observed over 100 days, the number of defective units is a random variable.

Using the daily data, a frequency distribution and relative frequencies can be calculated for each possible number of defects.

Defective units X	Frequency	Probability
0	5	0.05
1	25	0.25
2	40	0.40
3	25	0.25
4	5	0.05
Total	100	1.00

The table and the bar graph together describe the probability distribution of the number of defective units.

Figure label: Probability distribution of the number of defective units.

Types and Characteristics of Probability Distributions

- A discrete probability distribution gives the probability of each possible value and represents it as the height of a histogram bar.
- A continuous probability distribution represents probabilities for a continuous random variable X with a table or graph. It is shown as a curve.
- The mean of a probability distribution is called its expected value. In a histogram or density graph, probability is represented by area.
- Variance measures the average squared distance from the mean. For a discrete variable:

$$\text{Var}(X) = \sum X_i^2 P(X_i) - [E(X)]^2$$

Figure labels: discrete probability distribution; continuous probability distribution.

Types and Characteristics of Probability Distributions

- A discrete probability distribution shows the probability of each possible value as the height of a histogram bar.
- A continuous probability distribution describes a continuous random variable X using a table or graph and is drawn as a curve.

The slide compares the same examples of discrete and continuous random variables shown on Page 17.

Type	Example	Possible values
Discrete	One roll of a die	1, 2, 3, 4, 5, 6
Discrete	Number of employees	0, 1, 2, 3, 4, ...
Discrete	Heads in two coin tosses	0, 1, 2
Continuous	Battery usage time	$0 \leq X \leq 18$
Continuous	Student body weight	$35 \leq X \leq 120$
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Figure labels: discrete probability distribution; continuous probability distribution.

Probability Distribution of a Continuous Random Variable

1. Continuous random variable

If a random variable X can take every real value in a certain interval, X is called a continuous random variable.

2. Probability density function

Let X take every real value in $\alpha \leq X \leq \beta$. A function $f(x)$ defined on $\alpha \leq x \leq \beta$ is called the probability density function of X when all three conditions below are satisfied.

1. $f(x) \geq 0$

2. The area bounded by $y = f(x)$, the x -axis, $x = \alpha$, and $x = \beta$ is 1.

3. $P(a \leq X \leq b)$ equals the area bounded by $y = f(x)$, the x -axis, $x = a$, and $x = b$, where $\alpha \leq a \leq b \leq \beta$.

Normal Distribution

The normal distribution is one of the most widely used continuous probability distributions. Many datasets in the natural and social sciences have distributions that resemble a normal distribution.

Measurements such as height among people of the same sex and a similar age can form an approximately bell-shaped distribution that is relatively symmetric around the mean.

Example data: Korean Agency for Technology and Standards, 8th Korean Human Dimensions Survey

Average height: men 172.5 cm; women 159.6 cm

Graph labels: relative frequency; height (cm); men's mean 172.5 cm; women's mean 159.6 cm.

Source: Korean Agency for Technology and Standards / Size Korea, results of the 8th Korean Human Dimensions Survey (ages 20-69, surveyed 2020-2021). The graph is an explanatory normal-distribution model centered on the mean values.